Assignment 4: Reinforcement Learning

You are advising a team that wants to use a reinforcement learning agent to navigate an uncertain environment. The team has limited training resources and cares about reaching the goal, avoiding failures, and understanding when the agent's behaviour becomes unreliable.

Your task is to investigate two approaches:

- Q-learning: off-policy temporal-difference learning.
- PPO: on-policy actor–critic learning.

You must recommend an approach for specific operating conditions. Your recommendation must be supported by experiments, observations of individual episodes, and an explanation of the trade-offs.

The highest-scoring submission will make the most defensible recommendation, even if its agents perform poorly.

Following are your options:

1. MsPacMan
2. Breakout
3. SpaceInvaders

Part 1: Make predictions before training (30 points)

This is a mandatory step, if you do not submit this your main submission will not be considered → this is due on Friday the 13th of November

Before running the assessed experiments, submit a timestamped prediction sheet.

- For each statement below, predict what will happen and explain why:
- Which agent will improve fastest with a small interaction budget?
- Which agent will show the largest performance variation across seeds?
- How will stochastic transitions change the routes agents learn?
- How might a penalty for each step change success rate and episode length?
- Will the agent with the highest training return also perform best during evaluation?

- Which agent would you initially recommend, and what evidence would change your mind?

Use your team's option to identify three states where you expect the approaches to behave differently. For each state, record:

- The action you expect each agent to favour.
- The reason for your prediction.
- An observable result that would contradict it.

Your predictions are assessed for clarity and testability, not for being correct. Preserve the original sheet when your interpretation changes.

Part 2: Compare learning settings (20 Points)

Run the following investigation.

Setting	What changes?	Main question
A: Predictable environment	Deterministic transitions and terminal success reward	How efficiently does each approach learn?
B: Uncertain environment	Stochastic transitions, other conditions held constant	How does uncertainty affect learning and behaviour?
C: Changed incentives	Add a step penalty to Setting B	Does the reward encourage the behaviour people actually want?
D: Transfer	Evaluate Setting B agents on an unseen map, without further training	What survives a change in environment?

For Setting D, keep the observation and action formats compatible. Explain whether the state representation gives an agent a meaningful way to generalise. A tabular state ID may carry little transferable information when a map changes.

Required comparison rules

1. Choose and justify a common environment interaction budget. Count interactions, not just episodes.
2. Use the same training-seed list across algorithms where practical. Shared seeds do not guarantee identical experience once policies differ.
3. Keep evaluation separate from training and disable parameter updates during evaluation.

4. Use a separate evaluation-seed list and at least 200 evaluation episodes per trained agent.
5. Specify the evaluation policy: greedy Q-table actions, sampled policy actions, or highest-probability policy actions. Report any additional evaluation mode separately.
6. Give each approach the same hyperparameter-search allowance. Do not extensively tune one agent and leave the others at arbitrary defaults.
7. Keep held-out evaluation results out of tuning decisions.
8. Report runtime and hardware as well as interactions. Equal interactions do not imply equal computation.

Required evidence

For each condition, record:

- Success rate.
- Mean evaluation return.
- Failure rate.
- Episode length, with successful and failed episodes distinguished.
- Training interactions.
- Runtime.
- Variation across training seeds.

Show learning curves against environment interactions. State how curves are smoothed and how often evaluation occurs.

Do not pool all evaluation episodes across seeds and present them as independent training replications. Summarise results across trained agents.

When comparing Setting B with Setting C, use common outcome measures such as success and failure rate. Returns produced by different reward definitions are not directly comparable.

Part 3: Investigate the mechanisms (20 Points)

Change one main factor at a time and state what remains fixed.

Investigation	Practical experiment	Question to resolve
Exploration	Compare different epsilon schedules for Q-learning	How does behaviour during exploration affect the learned policy?
Planning horizon	Compare different discount factors	Which future outcomes does the agent effectively prioritise?

Policy exploration	Compare different PPO entropy coefficients	Does maintaining action diversity help learning here?
Update size	Compare different learning rates	When do updates become too slow or unstable?
PPO data reuse	Compare different numbers of update epochs per rollout	Does more optimisation of collected experience help within your budget?

Use at least five seeds for each tested configuration.

After running this experiment, go back to your submitted prediction. Then:

- Identify whether the result supports it.
- Examine a case that does not fit your explanation.
- Propose a plausible competing explanation.
- Run one small diagnostic experiment that could distinguish the explanations.

A diagnostic experiment should answer a question. Simply increasing the training budget is insufficient unless that specifically tests your explanation.

Part 4: Inspect what aggregate scores hide (20 Points)

Select ten recorded episodes from your runs:

- Three typical successes.
- Three typical failures.
- Two surprising success.
- Two surprising failure.

State how you selected them. Use the distribution of episode outcomes to justify which cases are “typical”; avoid selecting only striking examples.

For each episode, annotate three important decisions using the following structure:

Evidence	Your analysis
State and available actions	What choice did the agent face?
Chosen action and policy probabilities or Q-values	What does the agent’s current policy suggest?
Observed transition and outcome	What actually happened?
Alternative action	What might have happened, and what remains uncertain?

Distinguish a poor action from an unlucky outcome. A failed episode does not automatically demonstrate a bad policy.

Use the episodes to explain at least one limitation of your aggregate results.

Part 5: Conduct a human review (30 Points)

Review evidence within the team.

The review must:

1. Identify the strongest supporting evidence.
2. Identify one claim that goes beyond the evidence.
3. Propose one alternative explanation.
4. Compare this to the initial submitted predictions

Part 6: Make three recommendations (30 Points)

Recommend an agent for each situation:

Situation	Decision context
Rapid classroom demonstration	Training time is short and behaviour must be explainable.
Repeated operation in uncertainty	Failures matter more than taking additional steps.
Operation after an environment change	Reliability on unfamiliar conditions matters.

For each recommendation, provide:

- The chosen approach and configuration with justification.
- Two pieces of supporting experimental evidence.
- One observed weakness.
- A condition under which you would change your recommendation.
- One additional experiment needed before using it beyond this simulation.

You may conclude that none of the tested agents is adequately supported for a situation.

Explain where your findings concern the specific implementation and environment, and where you believe they illustrate a broader learning principle. Do not claim that one algorithm is universally superior.

What to submit

1. Executable notebook or scripts: One documented command or sequence that reproduces the main experiment.
2. Experiment records: Configurations, raw logs, seeds, software versions, and episode traces.
3. Prediction sheets: Original predictions and subsequent annotations.
4. Evidence brief: Maximum 2,000 words, excluding tables and figure captions, answering the questions above.
5. Review record: Maximum 1000 words

Every figure must identify its source runs. Another pair should be able to trace a plotted result back to its configuration and logs.

AI assistance (not graded but mandatory)

You may use AI for coding support, debugging, visualisation, and discussing concepts.

Include a short record of consequential AI assistance: what it suggested, what you accepted or rejected, and how you checked it. Routine autocomplete does not need to be logged.

You remain responsible for every submitted claim and result. Do not fabricate runs, predictions, peer feedback, or observations. Predictions reconstructed after seeing results do not count as prior predictions.